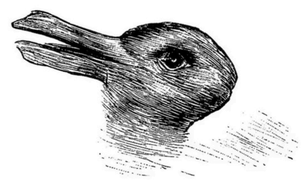
FizTeCh 2026

Redundancy of length in physics

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SUMMARY

Every measurement is, at its core, a comparison between quantities coincident in space and time; the local unit of length need therefore carry no intrinsic physical content. We render this redundancy explicit by extending the Hamiltonian phase space of general relativity with a compensating scale field, whereupon local rescalings become a gauge symmetry that leaves the algebra of observables unaltered. A change of conformal frame thus ceases to be a field redefinition and becomes a gauge transformation, and the choice of frame reduces to the choice of a clock that measures scales. Complete observables are invariant under this choice: an expanding universe with fixed masses and a static universe with growing masses yield identical measured redshifts, for example. Brans–Dicke theory of gravity serves as an illustrative case.

I The problem of frame

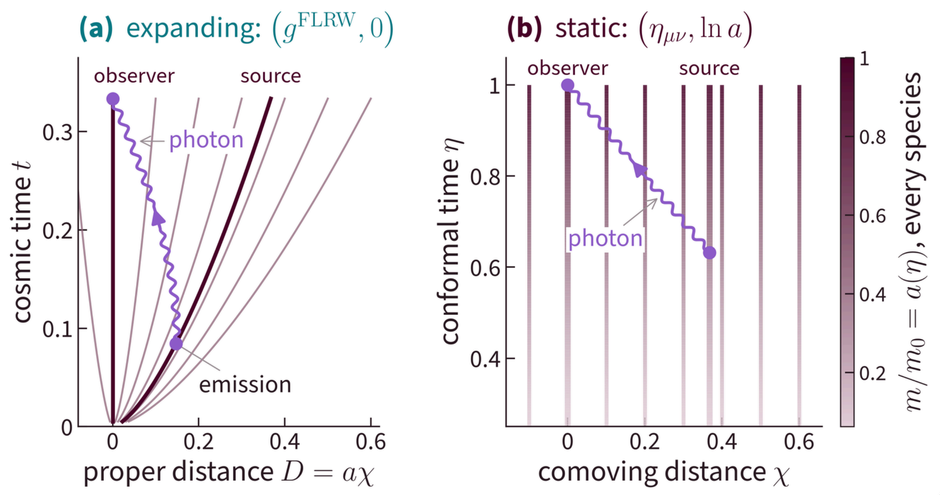
Partial observable: A physical quantity to which we associate a (measuring) procedure.

Complete observable: A quantity whose value (or probability distribution) can be predicted.

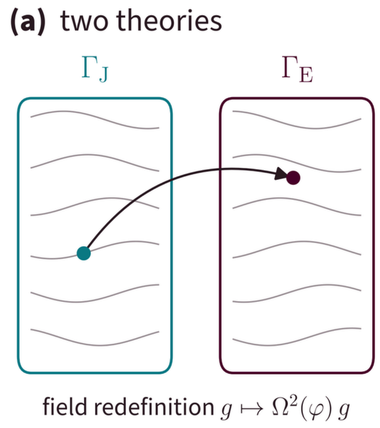
On a symplectic manifold (Γ_0, ω) with first class constraints $\mathfrak{C}_I$, the **complete observable** of f with respect to clocks T^I, is the value where the clocks read τ [1,2]:

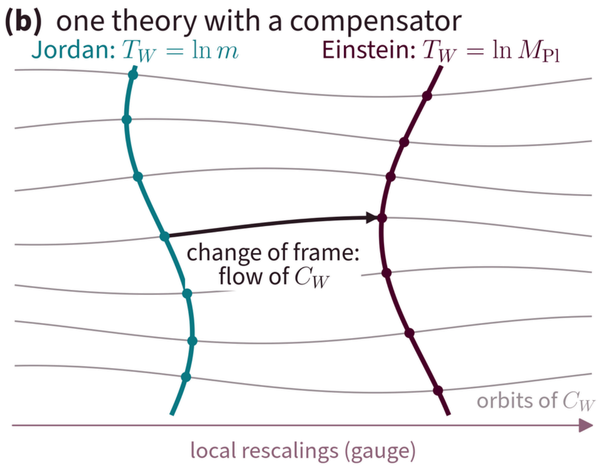
$$\mathrm{F}_{[f;T]}(\tau; y) = f(\mathrm{Fl}_{\varepsilon(y)}(y)), \;\; T^I(\mathrm{Fl}_{\varepsilon(y)}(y)) = \tau^I, \;\; \mathrm{F}_{[f;T]}(\tau) \approx \sum_n \frac{(\tau - T)^n}{n!} \{f, \tilde{\mathfrak{C}}\}_{(n)}$$

The observable is constant along gauge orbits; a change of frame is a field redefinition, crossing gauge orbits (see figure).



II Scale extension





We extend the phase space using a compensator σ into $\Gamma_{\mathrm{ext}} = \Gamma_0 \times \mathrm{T}^*(\mathcal{C}_\sigma)$ with the additional constraint $\mathfrak{C}_W = -\Pi_\sigma \approx 0$. For fields of scale weight w_k the undressed variables preserve the canonical form ⟶ change of chart is a symplectomorphism.

$$\{\mathfrak{C}_\parallel[\xi], \mathfrak{C}_\perp[N]\} = \mathfrak{C}_\perp[\mathcal{L}_\xi N] \qquad \{\mathfrak{C}_\perp[N], \mathfrak{C}_\perp[N']\} = \mathfrak{C}_\parallel[v] + \mathfrak{C}_W[\iota_v \mathrm{d}\sigma]$$
$$\{\mathfrak{C}_\parallel[\boldsymbol{\xi}], \mathfrak{C}_\parallel[\boldsymbol{\xi}']\} = \mathfrak{C}_\parallel[[\boldsymbol{\xi}, \boldsymbol{\xi}']] \qquad \{\mathfrak{C}_\perp[N], \mathfrak{C}_W[N]\} = -\mathfrak{C}_\perp[N\rho]$$
$$\{\mathfrak{C}_\parallel[\boldsymbol{\xi}], \mathfrak{C}_W[\rho]\} = \mathfrak{C}_W[\mathcal{L}_{\boldsymbol{\xi}}\rho] \qquad \{\mathfrak{C}_W[\rho], \mathfrak{C}_W[\rho']\} = 0$$

$\mathfrak{C}_W$ generates local rescalings: $\{\Phi, \mathfrak{C}_W[\rho]\} = w\rho\,\Phi, \;\; \{\sigma, \mathfrak{C}_W[\rho]\} = -\rho$

The reduced phase spaces are equivalent, since the scale adds no new observable.

$$\iota : \mathcal{A}_{\mathrm{D}}^{(0)} \mapsto \mathcal{A}_{\mathrm{D}}^{\mathrm{ext}} \longleftrightarrow \text{Isomorphic algebras}$$

III Frames are clocks

Scale clocks form an affine space over the weight-zero functions satisfying:

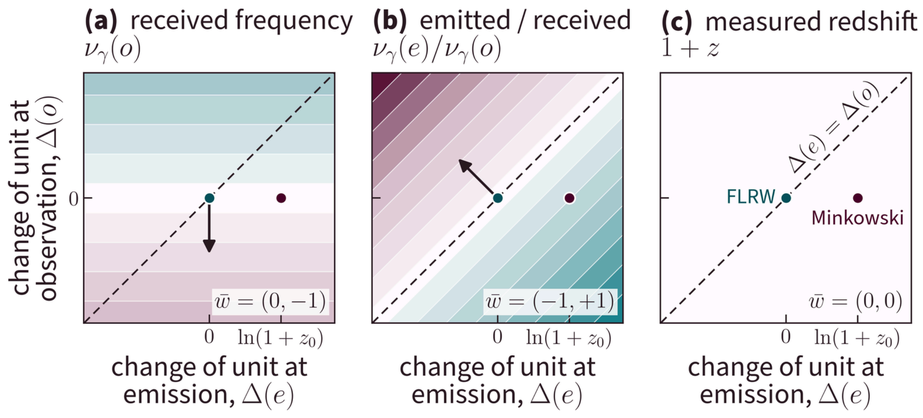
$$\{T_W, \mathfrak{C}_W[\rho]\} = -\rho$$

Choosing a clock σ, the partially complete constraints [3] are the dressed constraints

$$\mathrm{F}_{[\mathfrak{C}_\perp[N],\sigma]}(0) = \mathfrak{C}_\perp[N], \;\; \mathrm{F}_{[\mathfrak{C}_\parallel[\boldsymbol{\xi}],\sigma]}(0) = \mathfrak{C}_\parallel[\boldsymbol{\xi}], \;\; \mathrm{F}_{f;(T,T_W)}(\tau, 0) = \mathrm{F}_{\mathrm{F}_{f;T_W}(0);T}(\tau)$$

Observers measure products $\mathbf{R}$ of weighted local quantities, at relational events e, where the clocks take prescribed values, with event weights:

$$\bar{w}_e = \sum_{e_a = e} n_a w_a \qquad \mathbf{R} = \prod_a \Phi_a(e_a)^{n_a} \qquad \mathrm{F}_{[\Phi;T_W]}(\tau) = e^{w(T_W - \tau)}\Phi$$



IV Hamiltonian frame maps in Brans-Dicke theory

$$\mathrm{S} = \int \mathrm{d}^4x\sqrt{-g}\big[\frac{\phi}{2}R - \frac{\omega}{2\phi}(\partial\phi)^2 - V(\phi)\big] + \mathrm{S}_{\mathrm{matter}}[g, \chi]$$

$$\mathrm{S}_H = \int \mathrm{d}t\big[\int_S (\pi\,\dot{q} + p\,\dot{\chi} + p_\sigma\dot{\sigma}) - (\mathfrak{C}_\perp[N] + \mathfrak{C}_\parallel[\boldsymbol{\xi}] + \mathfrak{C}_W[\lambda])\big]$$

$$\delta X = \{X, \mathfrak{C}_W[\rho(t)]\} \longleftrightarrow (\delta N, \delta\boldsymbol{\xi}, \delta\lambda) = (\rho\, N,\, 0\,,\, \dot{\rho} - \mathcal{L}_{\boldsymbol{\xi}}\rho)$$

Choosing the clock $\phi = t$ the reduced Hamiltonian satisfies

$$h_E = h_J - \frac{\partial}{\partial\phi}\int_S \frac{Q_{ab}\pi^{ab}}{\phi}$$

Taking the lapse to be a canonical variable, the map generated by the constraint

$$\mathfrak{C}_W[\rho] + \int_S \rho\, N\pi_N \longrightarrow (\tilde{N}, \tilde{\pi}_N, \tilde{p}_\phi) = (\phi^{1/2}N, \phi^{-1/2}\pi_N, p_\phi - \frac{q_{ab}\pi^{ab}}{\phi} - \frac{N\pi_N}{2\phi})$$

is itself canonical, as long as we drop the last term

$$\{\tilde{N}(x), \tilde{p}_\phi(x')\} = \frac{N}{2}\phi^{-1/2}\delta(x, x') \neq 0$$

For $\omega = -\frac{3}{2}$ the term is no longer a problem because we get the primary constraint

$$\tilde{p}_\phi \approx 0$$

The Einstein and Jordan frames are related by a gauge fixing.

Which frame to use?

In general, frames differ physically when the coupling is not universal, the scale constraint is anomalous or $\phi \to \{0, \infty\}$.
Compute observables in a convenient frame and complete observables will match since they are scale invariant.

Conclusions

1. A conformal scale is an additional clock, measuring scale.
2. Complete observables are scale independent.
3. Scale independence is equal to interpretation invariance.

Outlook

1. Is the new constraint anomalous in quantum mechanics?
2. Causal structure and disformal maps.
3. Relaxing universality.

References

[1] C.Rovelli, Phys. Rev. D 65, 124013 (2002)
[2] B.Dittrich, Gen.Rel.Grav. 39 1891-1927 (2007)
[3] B.Dittrich, Quant.Grav. 23 6155-6184 (2006)